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Influence of space debris impact on solar array under power generation

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ABSTRACT

Recently solar array has become higher in potential and larger in capacitance. Therefore, possibility of collision between space debris and enlarged solar array has been pointed out. Many debris and dust impacts were confirmed on fuselage of the retrieved satellite Space Flyer Unit (SFU) and solar array paddle of satellite Eureca. If space debris collides with the solar array of an orbiting satellite, it may cause generation of high-density plasma by debris impact induced dielectric breakdown of satellite component and the phenomenon called sustained arc. By the short circuits on the solar array, current will not flow into a pay load of the satellite. And the worst event by this discharge is operational end of the satellite. However, any events of discharge phenomenon by debris impact cannot be yet confirmed. But we cannot ignore such possibility of discharge by debris impact. The purpose of this paper is to investigate discharge condition due to debris impact which yields us reduction of electric power of solar array, and to reduce influence of the impact on satellite missions. In this study, a solar array coupon was tested under hypervelocity impact in which a projectile was launched by a two-stage light gas gun installed in Kyushu Institute of Technology (KIT).

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1. Introduction

Recently, long duration operations spacecraft, higher in power, higher in potential, and the solar array especially higher in potential have been proposed for the actualization of large space platform for industrial use, such as the space factory, the space hotel, and solar power satellite. The use of high power in future space missions calls for high voltage power generation and transmission to minimize the energy loss and the cable mass. Satellites after their end of life, upper stages of rockets and the parts and fragments from them are called space debris. Solar arrays that are designed for long periods of operation are more likely to be impacted by space debris. The potential for impact is greater as the size of the satellites is larger. Collision of space debris with active solar arrays may cause generation of high-density plasma induced by impact. Then plasma grows up by surrounding plasma, and the phenomenon called discharge might take place. Space debris poses an obvious mechanical damage hazard to space assets, and may also precipitate a catastrophic electrical discharge that disrupts or disables onboard systems [1]. This discharge results in short circuits on the solar array and current does not flow into the satellite. This fact yields to the reduction of electric power of the solar array, and the impact influences on the satellite missions. Many debris and dust impacts were confirmed on fuselage of retrieved satellite SFU and solar array of satellite Eureca. Generation of the discharge phenomenon by debris impact is not yet confirmed, but such possibility will be increasingly important. For example, the discharge phenomenon called "sustained arc" is suggested as a cause of trouble of geostationary satellite Tempo-2.

2. Sustained arc experiment

2.1. Pseudo solar array coupon

In this study, a pseudo solar array coupon was used mainly because a real solar array coupon is expensive. The pseudo solar array coupon is shown in Fig. 1. In the pseudo solar array coupon, a copper plate was instead of a solar cell. The gap between the cells was approximately 0.6 mm. The real solar array coupon and the cross section are shown in Fig. 2. The gap between the cells was approximately 0.7 mm.

2.2. Experimental set-up and experimental condition

Hypervelocity impact tests in this study were conducted using a small two-stage light gas gun, installed at Laboratory of Spacecraft

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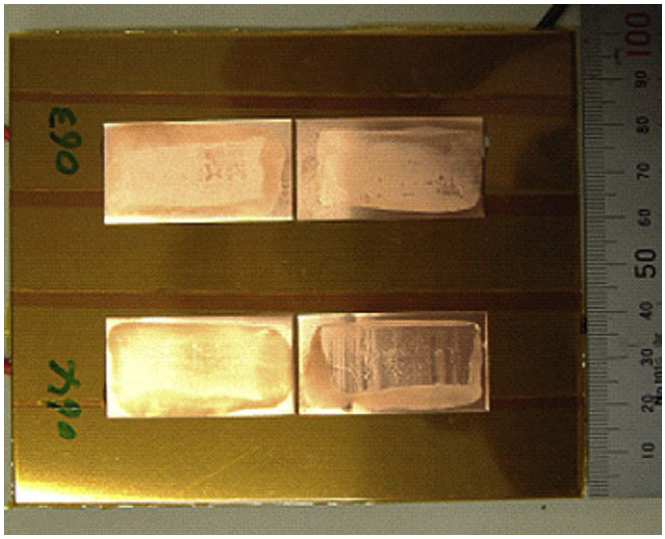

Fig. 1. Pseudo solar array coupon.

Environmental Interaction Engineering (LaSEINE) in Kyushu Institute of Technology shown in Fig. 3 [2]. Velocity range was 1.4–4.5 km/s, and the projectile was a 3 mm aluminum sphere (Al2024-T4) within a polycarbonate sabot. In the sabot separation section, sabot gets away from the aluminum sphere, and only the aluminum sphere collides on the pseudo solar array.

The pseudo solar array coupon of Fig. 1 was used as target and set up in a chamber. An external circuit that simulated the electric power of an actual solar array was connected to the coupon. In this study, impact velocity of the projectile, discharge current, and discharge voltage were measured by using four oscilloscopes. The current source used a current registry diode (CRD) with high-speed response. In the experiment, the voltage was set to 192 V and the current was set to 5 A. And voltage is imposed on a hot cell. The gap between the cells was approximately 0.6 mm. In this condition, when the projectile collides with coupon, the discharge is generated between cell and substrate or between cells. This phenomenon is called short circuit. At this time, the current flowing in this external circuit is measured by the three current probes (CP1–CP3) as shown in Fig. 4. The current probes CP1, CP2 and CP3 are employed to judge if short circuit on sustained arc happens between cells, and between cell and substrate, or not. The voltage probe (V) is employed to measure voltage change in the external circuit.

3. Plasma measurement and discharge test

In this section, we focus on plasma density, which may induce sustained arc.

3.1. Triple probe method and plasma measurement test

Triple probe [3] was set at a distance of 50 mm from the front of the pseudo solar array coupon to measure electron temperature and density of plasma created by hypervelocity impact. The configuration of the triple probe is shown in Fig. 5. The electron temperature T_e [eV] and density N_e [m^{-3}] can be calculated by V_2, V_3 and the following equations. The electron temperature is obtained from Eqs. (1) and (2);

$$\frac{I_1 + I_2}{I_1 + I_3} = \frac{1 - \exp(-\phi_{d2})}{1 - \exp(-\phi_{d3})} \tag{1}$$

$$I_1 = I_2 + I_3 \tag{2}$$

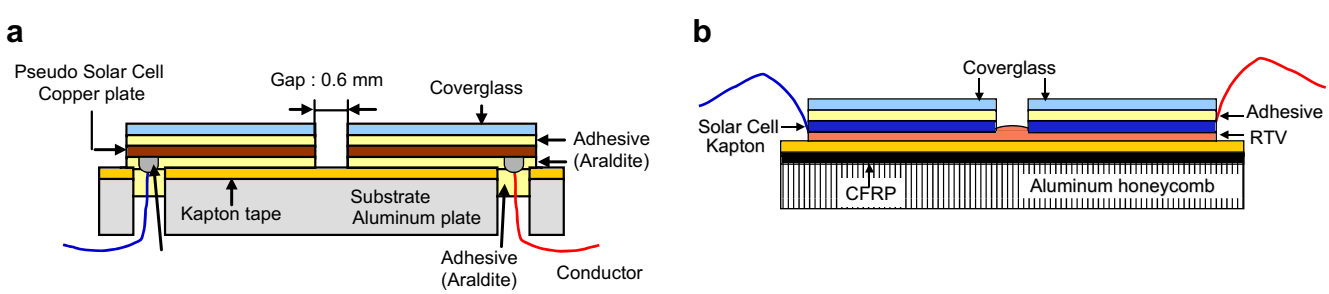

Fig. 2. Cross section of coupon. (a) Pseudo solar array coupon. (b) Real solar array coupon.

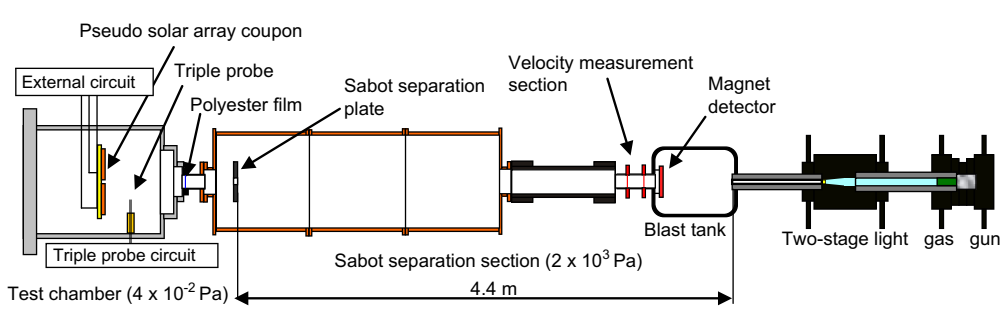

Fig. 3. Configuration of experimental setup.

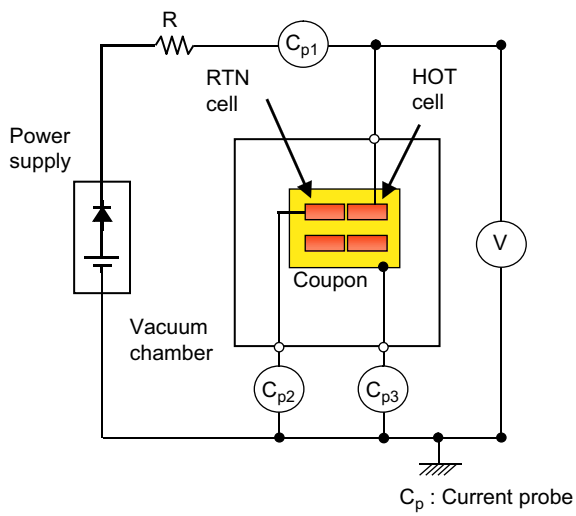

Fig. 4. External circuit for sustained arc experiment. CP1, CP2 and CP3 are probes to measure total current in the circuit, current flowing between cell 2 and substrate, current flowing between cells.

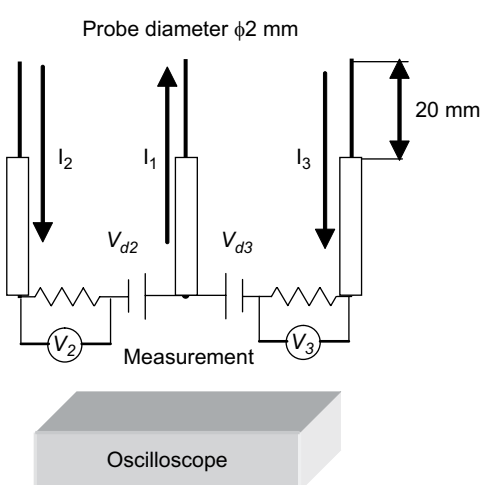

Fig. 5. Configuration of triple probe.

where $\varphi_{d2} = eV_{d2}/\kappa T_e$, $\varphi_{d3} = eV_{d3}/\kappa T_e$, e is elementary electric charge and κ is Boltzmann constant. Finally, electron density can be calculated by the following equations.

$$I_i = \frac{I_3 - I_2 \exp(-\phi_{\Delta V})}{1 - \exp(-\phi_{\Delta V})} \tag{3}$$

$$N_e = \left[(M)^{1/2}/S\right] I_i f_1(V_{d2}) \tag{4}$$

$$f_1(V_{d2}) = 1.05 \times 10^{15} (T_e)^{-1/2} [\exp(\phi_{d2}) - 1]^{-1} \tag{5}$$

Here M [g] is ion mass, S [mm^2] is surface area of probe and I_i [μA] is ion current.

Table 1 shows the experiment conditions and the results. In each experiment, the projectile was a 3 mm aluminum sphere (Al2024-T4) within a polycarbonate sabot. Fig. 6 shows the relationship between electron temperature and impact velocity. Electron temperature was approximately 2.5 eV. These electron temperatures are higher than that of plasma in low earth orbit, which is 0.09 eV.

Fig. 7 shows the relationship between electron density and impact velocity. This electron density was higher than that of plasma environment in low earth orbit, which is 1×10^{11} m^{-3}. Electron densities increase exponentially with impact velocity and this relationship is approximated by the following equation with impact velocity V [km/s].

$$N_e = 4 \times 10^{13} V^{6.4} \tag{6}$$

3.2. Discharge test

The pseudo solar array coupon after hypervelocity impact test is shown in Fig. 8. Impact velocity was 4.1 km/s. The projectile was a 3 mm aluminum sphere (Al2024-T4) within a polycarbonate sabot. In this experiment, it was done at some distance from the energized cell to prevent it from being destroyed.

Fig. 9 shows voltage waveforms of triple probe. Time is 0 when the external circuit for sustained arc experiment started discharge. Distance impact point between the triple probe was 72 mm. From Fig. 9, it was found that $V_{d1} = 2.25$ V, $V_{d2} = 15.6$ V, electron temperature and density calculated by Eqs. (1)–(5) were 3.5 eV and 3.6×10^{17} m^{-3}, respectively. It is higher density than low earth environment, where electron temperature and electron density are 0.09 eV and 1×10^{11} m^{-3} [4].

Figs. 10 and 11 show measured results of electron temperature and electron density including the plasma measurement experiment results (impact velocity: 1.5–2.5 km/s). In each experiment, the distance between triple probe and the impact point is not same, therefore the electron temperature is exchanged as the distance between triple probe and the impact point is 50 mm, assuming that the plasma spread as hemisphere. The ambient pressure in the vacuum chamber was approximately 10^{-4}–10^{2} Pa. The electron temperature and the electron density increase with impact velocity. Especially, the electron density increases exponentially.

The distance between the impact point and the HOT cell is 25 mm. From Fig. 13, the current (cell–substrate) decreased in 10–15 μs and just after the currents (cell-cell) started to flow. It is guessed that at first the discharge came off between the HOT cell and the impact point through the plasma created by hypervelocity impact, and after then, discharge pass changed to the HOT cell – the RTN cell from the HOT cell – the impact point because the RTN cell is nearer than the impact point. From Fig. 12, the discharge continued over 20 ms. But it did not develop to sustained arc. On

Table 1
Experimental conditions and the results

Case no.	1	2	3	4	5	6	7	8
Velocity [km/s]	1.43	1.56	1.92	2.13	2.16	2.23	2.27	2.50
Electron temperature T_e [eV] (50 mm)	2.9	2.0	2.2	2.5	2.3	2.6	2.7	3.5
Electron density N_e [m^{-3}] (50 mm)	4.2×10^{14}	8.3×10^{14}	1.6×10^{15}	7.6×10^{15}	1.3×10^{16}	9.6×10^{15}	6.7×10^{15}	9.1×10^{15}

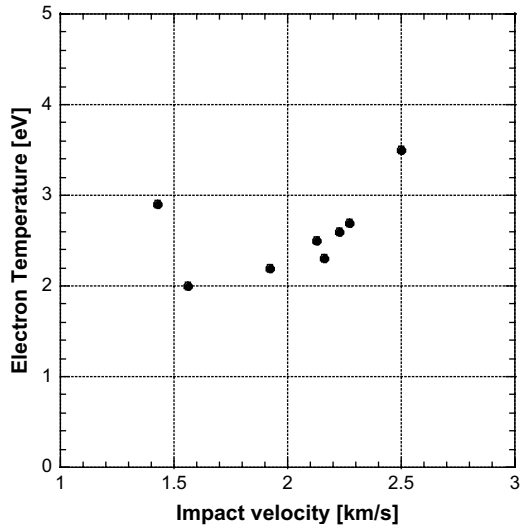

Fig. 6. Electron temperature.

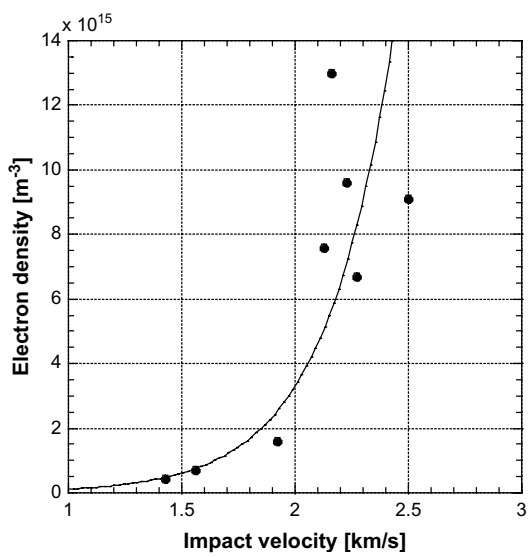

Fig. 7. Electron density.

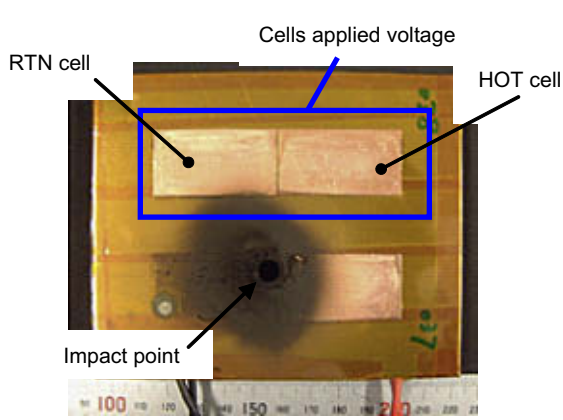

Fig. 8. Pseudo solar array coupon after impact test.

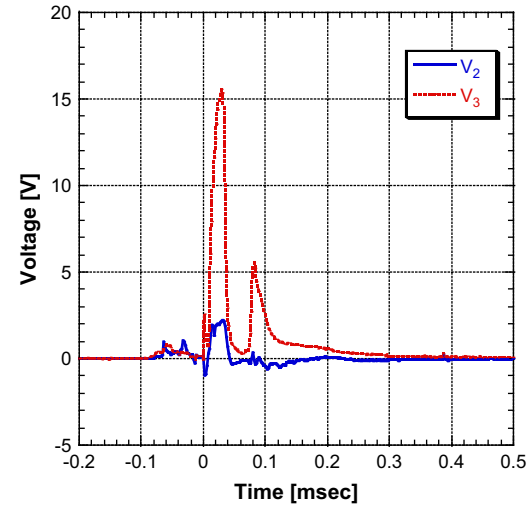

Fig. 9. Voltage waveforms of triple probe.

the pseudo solar array coupon after hypervelocity impact test, four discharge marks were found on the gap between HOT cell and RTN cell. Joule heating was generated as well as the insulation layer was broken.

4. Summary

In this study, hypervelocity impact tests were performed on solar array under pseudo power generation by an external circuit and collected data about short circuit between solar array and the substrate, in order to evaluate discharges and sustained arcs induced by space debris impact, the plasma created by hypervelocity impact was reasoned as well as the circuit current and the string voltage in the external circuit. In the results, it was determined the electron temperature and the electron density are

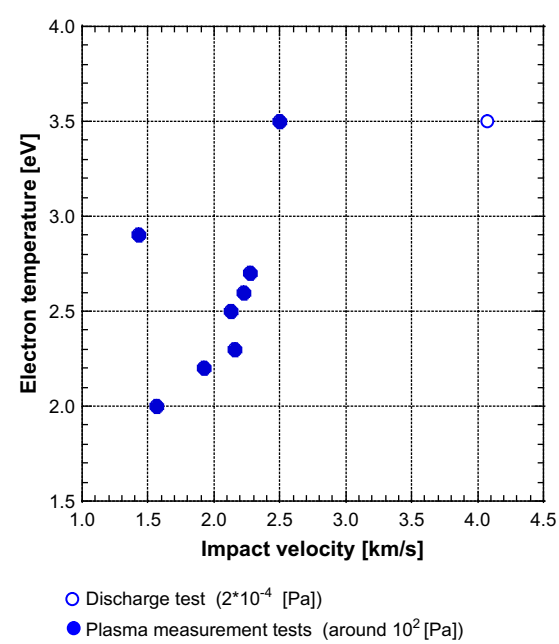

Fig. 10. Electron temperature.

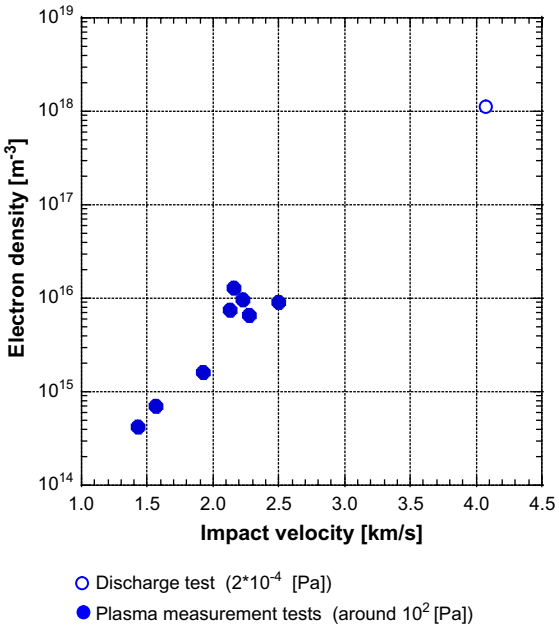

Fig. 11. Electron density.

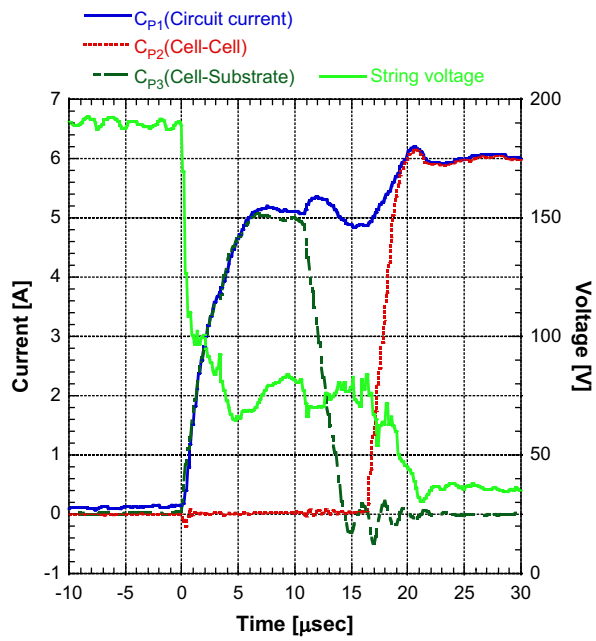

Fig. 13. Current and voltage of onset of arcing.

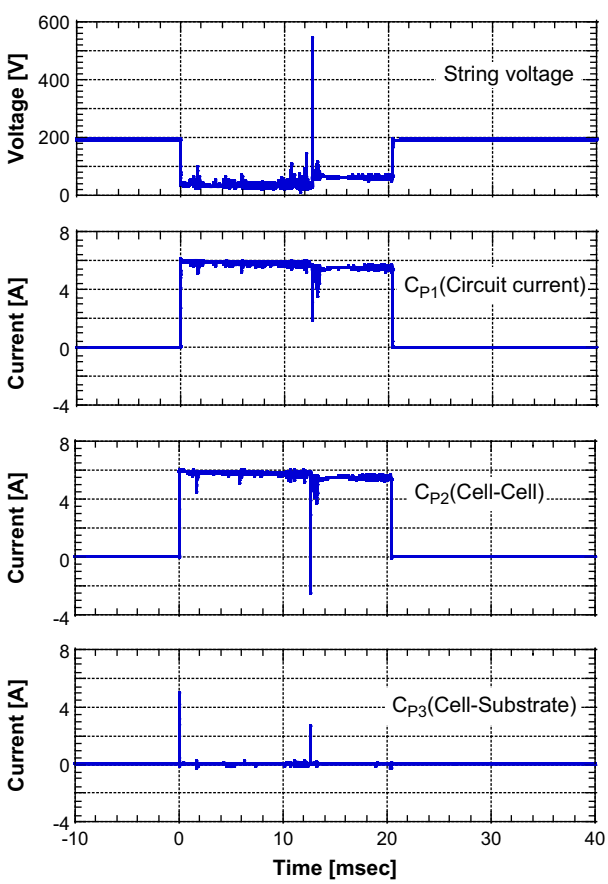

Fig. 12. Current and voltage waveforms.

function of the impact velocity. Especially, the electron density increases exponentially. Spread plasma created by hypervelocity impact can initiate discharge which is able to become sustained arc on detached point. We will carry out tests using a real solar array coupon in order to confirm sustained arc on the solar array coupon in the future.

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References

[1] Katz I, Davis VA, Snyder DB. Mechanism for spacecraft charging initiated destruction of solar arrays in GEO. In: AIAA Paper 98-1002, 36th Aerospace Science Meeting, Reno; 1998. 11 pp.
[2] Fukushige S, Akahoshi Y, Watanabe K, Harano T, Machida Y, Furusawa N, et al. Hypervelocity impact test to solar array for evaluation of possibility of sustained arc. In: The 25th international symposium on space technology and science, vol. 25; 2006. p. 56–1361.
[3] Chen S, Sekiguchi T. Instantaneous direct display system of plasma parameters by means of triple probe. Journal of Applied Physics 1965:2363–75.
[4] Tribble AC. The space environment implications for spacecraft design. Princeton University Press; 1995. p. 114.